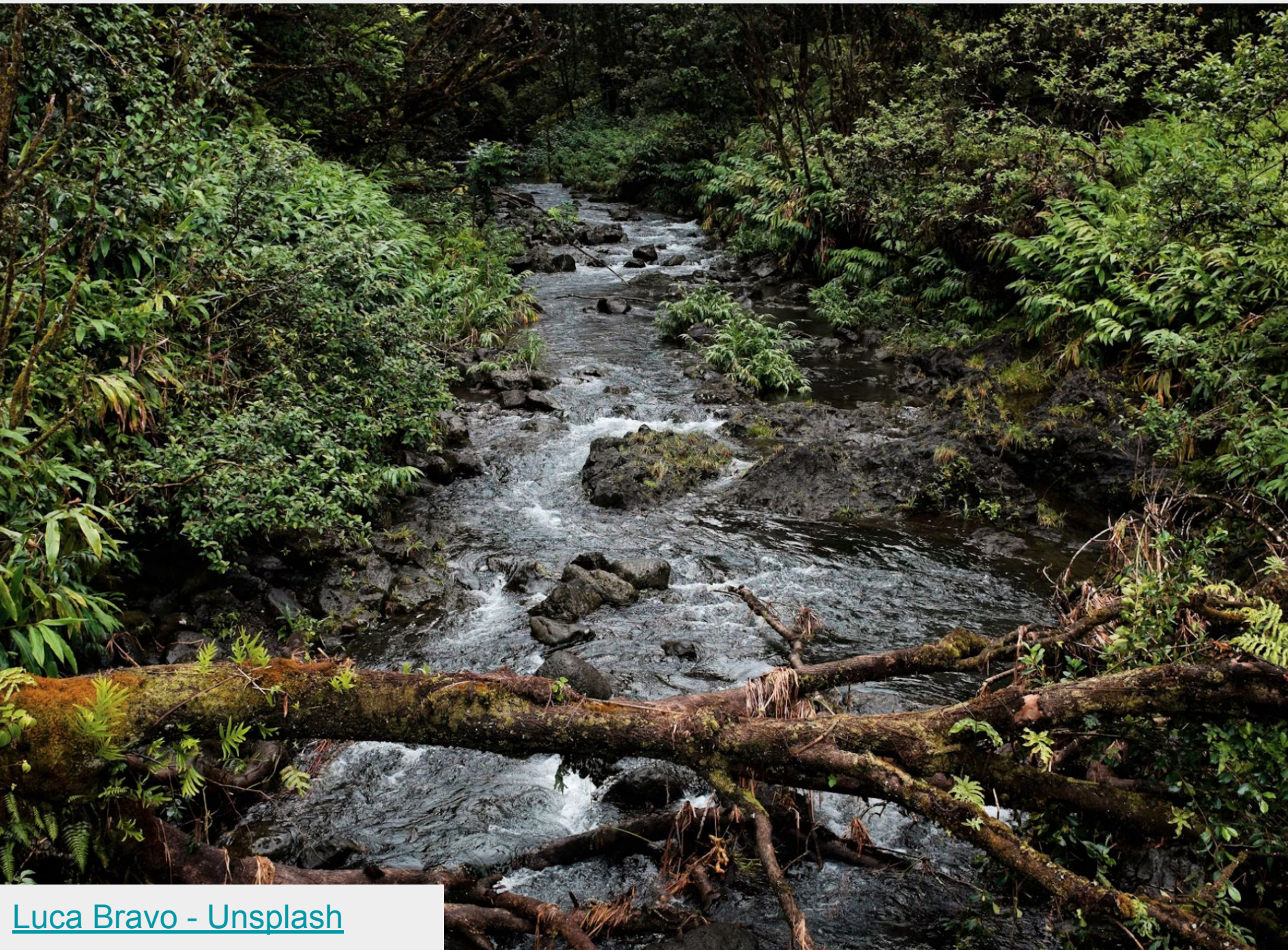
sed

Adapted from materials by Dr. Carrier



Luca Bravo - Unsplash

sed

Stream **ED**itor

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Stream can be a file or a piped input

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Stream can be a file or a piped input

General form:

```
sed [options] ‘/pattern/ command’ input_file
sed [options] ‘range command’ input_file
```

Common options

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`-E` to use extended regex

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`--in-place` or `-i` to modify a file in place

`--in-place=suffix`

Or

`-isuffix`

To create a backup file with suffix extension

Stream manipulation

Deleting lines:

- `sed '4 d'` - Deletes line 4
- `sed '1,4 d'` - Deletes lines [1,4]
- `sed -E '/pattern/ d'`

  Deletes lines that match the pattern

  ($ = end of file when using ranges)

Text substitution

Substituting text:

- `sed -E ‘s/pattern/replace_str/’`

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  ```
  sed -E ‘s/dog/cat/’ filename
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sed -E ‘s/ *$//’ filename
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- `sed -E 's/pattern/replace_str/'`
- Example:

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  sed -E 's/dog/cat/' filename
  ```

  ```
  sed -E 's/ *$//' filename
  ```

- By default, only replaces first match per line
  - Replace all:
    - `sed -E 's/pattern/replace_str/g'`
  - Replace second:
    - `sed -E 's/pattern/replace_str/2'`

Text substitution

Substituting text:

- `sed -E ‘s/pattern/replace_str/’`
- Example:

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sed -E ‘s/dog/cat/’ filename
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```
sed -E ‘s/ *$//’ filename
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Substituting text:

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- Example:

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sed -E 's/dog/cat/' filename
```

```
sed -E 's/ *$//' filename
```

- Can combine patterns with ranges:
- `sed -E '5,$ s/pattern/replace_str/'`

Text substitution

Substituting text:

- `sed -E ‘s/pattern/replace_str/’`
- Example:

```
sed -E ‘s/dog/cat/’ filename
```

```
sed -E ‘s/ *$//’ filename
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Text substitution

Substituting text:

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- Example:

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- We can also use parts of the matched pattern in our replacement! Uses (regex capture groups)

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sed -E 's/dog/cat/' filename
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sed -E 's/ *$//' filename
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- We can also use parts of the matched pattern in our replacement! Uses (regex capture groups)
- To refer to parentheses groups: `\1`, `\2`, …

```
sed -E 's/an? ([[:alpha:]]+) /the \1/'
```

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Substituting text:

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- Example:

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sed -E 's/dog/cat/' filename
```

```
sed -E 's/ *$//' filename
```

- We can also use parts of the matched pattern in our replacement! Uses (regex capture groups)
- To refer to parentheses groups: `\1`, `\2`, …

```
sed -E 's/an? ([[:alpha:]]+) /the \1/'
```

- To refer to the whole matched pattern: &

```
sed -E 's/.+\.com/www.&/'
```

Challenge questions

Use sed to normalize some text. Take several possible ways to represent "and", and convert them all to "and"

Examples:
And, AND, &, +, (more?) -> all replaced by and

How can we take a list of movies, and for any movie that is of the form:

"The XYZ"

we change it to

"XYZ, The"